# MECH 398 – MECHANICAL ENGINEERING LABORATORY I

Course Syllabus – Fall 2026


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This is your course syllabus. Please download the file and keep it for future reference.

# TEACHING TEAM

## Land Acknowledgement

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory. See: http://www.queensu.ca/encyclopedia/t/traditional-territories

## Inclusivity Statement

Queen’s students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: https://www.queensu.ca/vpcei/support/overview.

## Course Instructor

**Bradley Diak, PhD, PEng**
Professor
Mechanical and Materials Engineering
Queen’s University

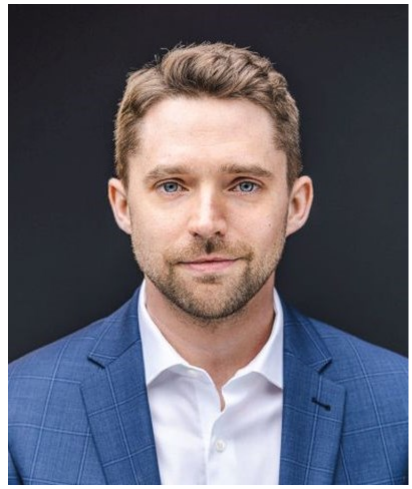
Office: Nicol Hall 302A
Phone:613-533-3180
E-mail: brad.diak@queensu.ca

## LABORATORY TECHNICAL SUPPORT

| | |
|---|---|
| **Gary Neff**<br>Mechanical Technologist<br>Mechanical and Materials Engineering<br>Queen's University<br><br>Office: McLaughlin Hall, Room 215B<br>Phone:613-533-6000 X78885<br>E-mail: gary.neff@queensu.ca |  |
| **Adam Cordick**<br>Electronics Technologist<br>Mechanical and Materials Engineering<br>Queen's University<br><br>Office: McLaughlin Hall, Room 215<br>Phone:613-533-6000 X78380<br>E-mail: adam.cordick@queensu.ca |  |

## LAB LECTURERS

| | |
|---|---|
| **REFRIGERATION**<br><br>**Jackson Crane, PhD, PEng**<br>Assistant Professor<br>Mechanical and Materials Engineering<br>Queen's University<br><br>Office: McLaughlin Hall, Room 303<br>Phone:613-533-6000<br>E-mail: jackson.crane@queensu.ca |  |

| | |
|---|---|
| **FLOW in PIPES**<br><br>**John Kurelek, PhD**<br>Assistant Professor<br>Mechanical and Materials Engineering<br>Queen's University<br><br>Office: McLaughlin Hall, Room 208<br>Phone:613-533-6000, 76258<br>E-mail: john.kurelek@queensu.ca |  |
| **VERTICAL JUMP HEIGHT**<br><br>**Qingguo Li, PhD, PEng**<br>Professor<br>Mechanical and Materials Engineering<br>Queen's University<br><br>Office: McLaughlin Hall, Room 301A<br>Phone:613-533-3191<br>E-mail: ql3@queensu.ca |  |
| **STRUCTURES**<br><br>**Yong Jun Lai, PhD, PEng**<br>Professor<br>Mechanical and Materials Engineering<br>Queen's University<br><br>Office: McLaughlin Hall, Room 226<br>Phone:613-533-6535<br>E-mail: lai@queensu.ca |  |

## TEACHER ASSISTANT INFORMATION – PLEASE SEE THE COURSE ONQ.

# MECH 398 (F K2)

## Course Description

This is the first of two laboratory courses in the third year of the General Option of the Mechanical Engineering program. Lecture topics are selected to provide the background required to undertake the laboratory work.

(0/0/0/24/0) (Mathematics/Natural Sciences/Complementary Studies/Engineering Science/Engineering Design)

## Mode of Instruction

The delivery mode of this course is blended.

## Course Location and Time

Please refer to the onQ course page.

## Pre-Requisite Knowledge

This course is designed for learners who have completed second year Mechanical Engineering who have basic knowledge in fluid mechanics, thermodynamics, solid mechanics, vibrations and heat transfer. Instrumentation, data acquisition, and data analysis are covered in the course.

## Course Learning Outcomes (CLO)

By the end of this course, students should be able to:

| CLO | Description | Indicator[1] |
|---|---|---|
| CLO 1 | Use experimental apparatus specific to Mechanical Engineering | IN-Conduct |
| CLO 2 | Apply theoretical concepts to physical and simulated systems | IN-Analysis |
| CLO 3 | Apply critical thinking to solve in-lab problems | IN-Synthesis |
| CLO 4 | Efficiently write effective technical lab reports following academic integrity principles | CO-Written<br>EE-AI |
| CLO 5 | Work effectively and safely in teams | TW-Contribution<br>TW-Feedback<br>IN-Safety<br>CO-Spoken |
| CLO 6 | Submit reports according to a deadline | PR-Interpersonal<br>EE-AI |

[1] For further information on graduate attributes and indicators, please click here.

## Course Evaluation

### Assessment Weighting

| Assessment Tool | Due Date (before 23:59 ET) | Weight | Alignment with CLOs |
|---|---|---|---|
| Tutorials | By 23h59 of the tutorial date. | **4x3% = 12%** | 2, 3, 5, 6 |
| Pre-Lab Quizzes | Before lab session. | **4x2% = 8%** | 1, 2, 3, 5, 6 |
| In-Lab TA Assessment | At end of lab session. | **4x2% = 8%** | 5 |
| Virtual Team Notebook | All notes from individual's and their teams activities in the course are added into one virtual document (the notebook). A TA sign-off is required for entries made during the lab session at the end of the activity. The final curated notebook is due within 5 days of the last lab report submission. | **8%** | 1, 4, 5 |
| Lab Reports | 10 days after completion of the lab activity by 23h59. | **4x16% = 64%** | 3, 4, 6 |
| | | **100%** | |

## Assessment Descriptions

### Tutorials

There is one tutorial assignment for each of the four labs that must be completed for each of the four labs. The assignments and the method of handing in the assignments vary with each lab. Some labs have verification quizzes that must be completed in the tutorial time.

### Pre-Lab Quizzes

There are four pre-lab quizzes in the course, one for each lab. The quizzes are comprised of multiple choice, true/false and matching response questions that will help to identify gaps in conceptual understanding and will prepare students for the lab activity. These quizzes are taken on the class website in onQ. You have a 24 hour window of opportunity to start the quiz, but it must be completed before the lab starts. Once the quiz is started, it remains open until you complete it, or until your lab session starts, whichever comes first.

### In-lab Assessment

At the end of each remote lab session, the TA assigns a grade to the lab Team based on their effectiveness in doing the lab activity. This is a team mark, but anyone not engaged and contributing with their team can be down-marked separately from the team grade.

### Notebook

Notes are taken for individual and team-based activities in the course, including lectures, tutorials, lab sessions and any course thinking outside the lab. Good notetaking enables fast and efficient data compilation, analysis and report writing. A notebook is a living diary of your activities. In lieu of a permanently bound hard-cover notebook, learners will use a cloud-based MSWord document shared with their assigned team members and TAs only. An organized document compiling all team members' entries is the assessment deliverable due within five days of submission of the final lab report. See the onQ marking rubric for requirements.

### Lab Reports

Each lab requires submission of an individual lab report. Each lab report is due at midnight 10 days after the completion of the lab activity. Refer to the report MSWord template on onQ.

## Grading

All assessments in this course will receive numerical percentage marks. The final grade you receive for the course will be derived by converting your numerical course average to a letter grade according to the established Grade Point Index.

### Feedback on Assessments

Feedback on graded activities will be made via onQ. You can expect feedback on your lab reports within 18 days of the due date.

### Accessing Your Final Grade

The onQ gradebook can be used to track your grades in real-time. Your final grades will show up on SOLUS. Official transcripts showing final grades will be available on the Official Grade Release Date. Please note that in official transcripts, a mark of IN (incomplete) is considered a grade, and your transcript is released with this grade.

## Course Materials

### Required Textbook

There is no required textbook for this course.

## Other Materials

All other course material is accessible via OnQ.  Safety glasses are mandatory to wear in all labs.  Please provide your own safety glasses.

## Required Hardware/Software

Students must have a reliable internet connection and hardware that are compatible with online learning requirements.   A computer with typical software for writing (Text Editor), data analysis (Excel, Matlab, Python) and communication (Zoom and Teams) is required.

All other course material is accessible via the course onQ.

## Suggested Time Commitment

This course represents a study period of one semester (12 weeks). There are four separate lab activities that run in consecutive three-week blocks, with each lab activity in a block taking about 2 weeks and consisting of: lecture, tutorial, pre-lab quiz, lab activity, submit report. Learners who adhere to a pre-determined study schedule are more likely to successfully complete the course.

## Lab Learning Outcomes

| Lab | Lab Learning Outcomes |
|---|---|

**Refrigeration**

By the end of this topic, learners will be able to:

- [LLO 1] Describe the use of a refrigeration cycle [CLO 2, 4]
- [LLO 2] Describe and understand the vapour compression cycle [CLO 2, 3, 4]
- [LLO 3] Effectively lower the water temperature using a refrigeration cycle [CLO 1,2,3,5]
- [LLO 4] Explain how the water mass flow rate affects the system parameters such as the coefficient of performance [CLO 1, 2, 3, 4]

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**Flow in Pipes**

By the end of this topic, learners will be able to:

- [LLO 1] Explain the characteristics of a flow in a piping system [CLO 2, 4]
- [LLO 2] Measure losses in a piping system [CLO 1, 5]
- [LLO 3] Apply Bernoulli's Equation with losses to flow in a pipe [CLO 2, 3]
- [LLO 4] Calculate pressure drops, exit velocity profile, Reynolds number, volume flow rate, friction factors and loss coefficients [CLO 2, 3, 4]
- [LLO 5] Compare predictions and measurements [CLO 3, 4]
- [LLO 6] Instrument the piping system with manometers [CLO 1]

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**Measuring and Predicting Vertical Jump Height**

By the end of this topic, learners will be able to:

- [LLO 1] Apply free body diagrams to a dynamic system [CLO 2, 3]
- [LLO 2] Approximate vertical jump height by applying the Flight Time and the Impulse-Momentum methods [CLO 2, 3, 5]
- [LLO 3] Compare the effects of the countermovement and squat jumps on loading of the ankle joint and jump height [CLO 1, 3]
- [LLO 4] Explain the importance of system calibration [CLO 1, 3]
- [LLO 5] Use Matlab or Python codes to model physical systems [CLO 1]

| Lab | Lab Learning Outcomes continued |
| --- | --- |
| **Structures** | By the end of this topic, learners will be able to:<br>• [LLO 1] Explain the correlation between stress and strain [CLO 2, 3, 4]<br>• [LLO 2] Quantitatively observe the differences between stress and strain and their magnitudes [CLO 2,3]<br>• [LLO 3] Recognize that the fundamental calculations and understanding of stress and strain is necessary to understand CAD analysis [CLO 2]<br>• [LLO 4] Recognize where to place strain gauges on a test specimen [CLO 1, 2, 3]<br>• [LLO 5] Explain the difference between axial and hoop stress, and how to calculate them [CLO 2, 3]<br>• [LLO 6] Transform strain components [CLO 2,3]<br>• [LLO 7] Identify stiffer and compliant directions in a geometrical lattice [CLO 2, 3] |

# COURSE COMMUNICATION

## Questions About Course Material

Questions or comments regarding the course material that can be of benefit to other students should be posted in the Q&A forum on the class website for each specific lab. The instructor, TAs, and students are encouraged to answer these questions directly in the discussion forum for the benefit of everyone in the course.

## Course Announcements

The instructor will routinely post course news in the Announcements section on the main course homepage on OnQ. Please sign up to be automatically notified by email when the instructor posts new information in the Announcements section. Instructions on how to modify your notifications are found in the Begin Here section of the onQ course site.

## Office Hours

In addition to interaction in the Q&A discussion forums, you will have the opportunity to interact with either a TA or the instructor during office hours. The instructor will provide a schedule of availability at the beginning of the term.

## Confidential Matters

If you have a confidential matter you would like to discuss with the Teaching Team, their contact details are on the first page of this document. Expect e-mail replies within 24 hours.

### Absences (Academic Considerations) and missed assignments

Students who miss assignments, quizzes, classes, labs, tutorials or other course deliverables due to unforeseen personal circumstances outside of their control can apply for Academic Consideration. For information on academic considerations due to extenuating circumstances, please review the information on the Smith Engineering website. Note that unacceptable reasons include malfunctioning computer, extracurricular activities, work, travel plans, scheduling conflicts, generally being behind on schoolwork, etc. Do not schedule travel during midterms and final exams, as travel and work are not acceptable reasons for granting academic considerations.

**Please email processed academic considerations to the Course Coordinator: diak@queensu.ca.**

### Late Submission Policy

Tutorials are due by 23h59 the day of the tutorial but can be submitted up to one day late with a penalty of 50% after which the drop box is closed. Attendance at scheduled lab sessions is mandatory. Reports submitted for a lab that was not attended will be penalized 50%. Arriving more than 10 minutes late for a lab session is treated as an absence unless you have a valid explanation. All deliverables have fixed due dates but can be submitted late with penalty. Any applicable late penalties are described in the details for each assessment. For example, late submissions of lab reports will incur a 10% penalty per day until a score of zero. In the event of extenuating circumstances, you must follow the policies for requesting an academic consideration (as described above). In the absence of an approved consideration request, the normal late penalty will apply as described in the assignment or any course/departmental policies.

## STANDARD QUEEN'S AND SMITH ENGINEERING POLICIES

### Netiquette

In this course, you may be expected to communicate with your peers and the teaching team through electronic communication. You are expected to use the utmost respect in your dealings with your colleagues or when participating in activities, discussions, and online communication.

The following is a list of netiquette guidelines. Please read them carefully and use them to guide your online communication in this course and beyond.

1. Make a personal commitment to learn about, understand, and support your peers.
2. Assume the best of others and expect the best of them.
3. Acknowledge the impact of oppression on the lives of other people and make sure your writing is respectful and inclusive.
4. Recognize and value the experiences, abilities, and knowledge each person brings.

5. Pay close attention to what your peers write before you respond. Think through and re-read your writings before you post or send them to others.
6. It's alright to disagree with ideas, but do not make personal attacks.
7. Be open to be challenged or confronted on your ideas and challenge others with the intent of facilitating growth. Do not demean or embarrass others.
8. Encourage others to develop and share their ideas.

## STUDENT CODE OF CONDUCT

Queen's University values maintaining an environment free of, and will not tolerate, harassment, discrimination, and reprisal. The Student Code of Conduct applies to all students at Queen's. It outlines the activities and behaviours that could be considered Non-Academic Misconduct (NAM). The Code also describes the NAM process and the sanctions that could be imposed on a student found responsible for a violation. All students should be familiar with the Student code of conduct and related policies on sexual violence prevention and response and harassment and discrimination prevention and response.
https://www.queensu.ca/nonacademicmisconduct/policies

## COPYRIGHT

Course materials created by the course instructor, including all slides, presentations, synchronous and asynchronous course recordings, handouts, tests, exams, and other similar course materials, are the intellectual property of the instructor. It is a departure from academic integrity to distribute, publicly post, sell or otherwise disseminate an instructor's course materials or to provide an instructor's course materials to anyone else for distribution, posting, sale or other means of dissemination, without the instructor's ***express consent***. A student who engages in such conduct may be subject to penalty for a departure from academic integrity and may also face adverse legal consequences for infringement of intellectual property rights and, with respect to recordings, potentially privacy violations of other students.

## ACADEMIC INTEGRITY

As an engineering student, you have made a decision to join us in the profession of engineering, a long-respected profession with high standards of behaviour. As future engineers, we expect you to behave with integrity at all times. Please note that Engineers have a duty to:

•Act at all times with devotion to the high ideals of personal honour and professional integrity.

•Give proper credit for engineering work

The standard of behaviour expected of professional engineers is explained in the Professional Engineers Ontario Code of Ethics. Information on policies concerning academic integrity is available in the Queen's University Code of Conduct, in the Senate Academic Integrity Policy Statement, on the Smith Engineering website, and from your instructor.

Departures from academic integrity include plagiarism, use of unauthorized materials or services, facilitation, forgery, falsification, unauthorized use of intellectual property, and collaboration, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the University.

In the case of online or remotely proctored exams, impersonating another student, copying from another student, making information available to another student about the exam questions or possible answers, posting materials to online services, communicating with another person during an exam or about an exam during the exam window, or accessing unauthorized materials, including internet sources and using unauthorized materials, including smart devices, are actions in contravention of academic integrity.

## Plagiarism/Use of Generative Artificial Intelligence (GenAI) Tools:

Students must submit their own work and cite any work that is not theirs. The Generative AI tool LibreChat may be permissible only if explicitly noted in the assignment instructions. In these cases, the material must be cited, and all cited material must be retrievable in exactly the same form as created, that is perpetually constant as one would expect from referencing a printed library book. Any other use of GenAI tools in this course constitutes a Departure from Academic Integrity as defined by University Academic Integrity procedures.

## Turnitin statement

This course makes use of Turnitin, a third-party application that helps maintain standards of excellence in academic integrity. Normally, students will be required to submit their course assignments through onQ to Turnitin. In doing so, students' work will be included as source documents in the Turnitin reference database, where they will be used solely for the purpose of detecting plagiarized text in this course. Data from submissions is also collected and analyzed by Turnitin for detecting Artificial Intelligence (AI)-generated text. These results are not reported to your instructor at this time but could be in the future.

Turnitin is a suite of tools that provide instructors with information about the authenticity of submitted work and facilitates the process of grading. The similarity report generated after an assignment file is submitted produces a similarity score for each assignment. A similarity score is the percentage of writing that is similar to content found on the internet or the Turnitin extensive database of content. Turnitin does not determine if an instance of plagiarism has occurred. Instead, it gives instructors the information they need to determine the authenticity of work as a part of a larger process.

Please read Turnitin's Privacy Policy, Acceptable Use Policy and End-User License Agreement, which govern users' relationship with Turnitin. Also, please note that Turnitin uses cookies and other tracking technologies; however, in its service contract with Queen's Turnitin has agreed that neither Turnitin nor its third-party partners will use data collected through cookies or other tracking technologies for marketing or advertising purposes.

For further information about how you can exercise control over cookies, see Turnitin's Privacy Policy.

Turnitin may provide other services that are not connected to the purpose for which Queen's University has engaged Turnitin. Your independent use of Turnitin's other services is subject solely to Turnitin's Terms of Service and Privacy Policy, and Queen's University has no liability for any independent interaction you choose to have with Turnitin.

*Portions of this document have been adapted, with permission, from the University of Toronto Centre for Teaching Support and Innovation tip sheet "Turnitin: An Electronic Resource to Deter Plagiarism".*

# ACADEMIC AND STUDENT SUPPORT

Queen's has a robust set of supports available to you, including the Library, Student Academic Success Services (Learning Strategies and Writing Centre), and Career Services. Learners are encouraged to visit the Smith Engineering Current Students web portal for information about various other policies such as academic advisors, registration, student exchanges, awards and scholarships, etc. Students are also encouraged to review the information that is available in the EngQ Hub, posted in onQ.

## ABSENCES - ACADEMIC CONSIDERATIONS

Academic Consideration is a process by which students request validated time away from their academics when they are experiencing an extenuating circumstance or personal circumstances beyond a student's control that interfere with their ability to complete academic requirements.

For additional information and to apply for academic consideration, please review the information on the Smith Engineering's Academic Consideration (Absences) website.

## ACADEMIC ACCOMMODATIONS

Academic Accommodations are available when students need an adjustment to their learning environment due to a reason related to a Human Rights protected ground. Below there is information on the most common Academic Accommodations requested by students. For more information, please see Smith Engineering's Academic Accommodation website.

## ABSENCES – REQUESTS FOR EXCUSED ABSENCES FOR SIGNIFICANT EVENTS (REASE)

Students participating in a Varsity Athletics event, the student reserve forces, a non-varsity event at the provincial, national, or international level, or an event to which you were invited as a distinguished guest can submit a Request for Excused Absence for Significant Event following the REASE process outlined on the Student Affairs webpage.

Students participating in Design Competitions supported by Smith Engineering can submit a Request for Excused Absence for Significant Events (REASE) following the process outlined on the Smith Engineering's Academic Consideration (Absences) website.

All Requests for Excused Absences for Significant Events must be submitted to the Faculty a minimum of 2 weeks prior to the event start date.

## RELIGIOUS ACCOMMODATIONS

Students in need of accommodation for religious observance are asked to complete the Religious Observance and Academics Activities form within a week of receiving their course syllabus (each term) and submit the form via email to the Engineering Program Advisor, (Accommodations & Considerations), engineering.aac@queensu.ca

- Alternative assignments are considered a "reasonable accommodation" under the Ontario Human Rights Code
- Please refer to the Queen's Multi-faith Calendar for all of this year's faith dates
- Students with questions about their rights and responsibilities regarding religious accommodations can contact chaplain@queensu.ca

## OTHER HUMAN-RIGHTS BASED ACCOMMODATIONS

Engineering students who need accommodation on Human Rights grounds are asked to contact, in confidence, the Engineering Program Advisor, (Accommodations & Considerations) engineering.aac@queensu.ca or to call 613-533-6000 x 78013.

## ACCOMMODATIONS FOR DISABILITIES

Queen's University is committed to working with students with disabilities to remove barriers to their academic goals. Queen's Student Accessibility Services (QSAS), students with disabilities, instructors, and faculty staff work together to provide and implement academic accommodations designed to allow students with disabilities equitable access to all course material (including in-class as well as exams). If you are a student currently experiencing barriers to your academics due to disability related reasons, and you would like to understand whether academic accommodations could support the removal of those barriers, please visit the QSAS website to learn more about academic accommodation.

To start the registration process with QSAS, click the "**Access Ventus"** button found on the Ventus student portal.

Ventus is an online portal that connects students, instructors, Queen's Student Accessibility Services, the Exam's Office, and other support services in the process to request, assess, and implement academic accommodations. To learn more about Ventus, visit A Visual Guide to Ventus for Students:
https://www.queensu.ca/ventus-support/students/visual-guide-ventus-students

For questions or assistance with requesting Academic Consideration or Accommodation, contact the Smith Engineering Program Advisor (Accommodations and Considerations) at engineering.aac@queensu.ca

Every effort has been made to provide course materials that are accessible. For further information on accessibility compliance of the educational technologies used in this course, please consult the links below.

## Student Wellbeing

Success in Engineering includes taking care of your wellbeing. If you're experiencing stress, anxiety, burnout, personal challenges, or you're simply not sure where to turn, EngWell is here to help.

EngWell provides confidential support through our Wellness Advisor, Embedded Counsellors, mental health programming, and referrals to academic and campus resources. Whether you need short term guidance, help navigating university supports, or someone to talk to, we encourage you to reach out early.

## Educational Technology

| Educational Technology | Accessibility Compliance Information |
|---|---|
| onQ<br>(Brightspace Learning Management System by D2L) | https://www.d2l.com/accessibility/standards/ |
| MS-Teams | https://support.microsoft.com/en-us/office/accessibility-support-for-microsoft-teams-d12ee53f-d15f-445e-be8d-f0ba2c5ee68f |
| Zoom | https://zoom.us/accessibility |

If you find any element of this course difficult to access, please discuss with your instructor how you can obtain an accommodation.

## Technical Support

Some basic comfort levels with basic hardware and software skills are required for this course. If you require technical assistance, please contact Technical Support.